NIOS Class 10 (211) Mathematics — Guaranteed Sure-Shot Questions & Solved Templates

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

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> **These 6 question templates appear in almost every single NIOS exam paper.**
> Mastering these 6 templates alone locks in **25 to 30 marks** in Section B! Practice writing each of these out by hand.

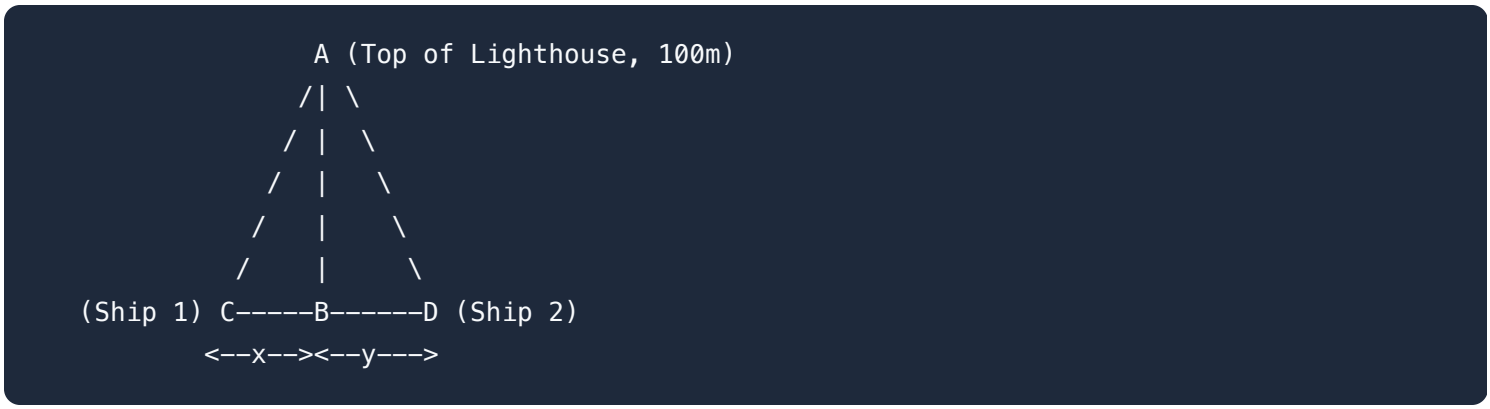
Template 1: Heights and Distances (5 Marks — Q44)

Problem (Real Exam Question from 2024 & 2025 Papers):

> As observed from the top of a $100\ \text{m}$ high lighthouse from the sea level, the angles of depression of two ships on the opposite sides of the lighthouse are 60° and 45° respectively. Find the distance between the two ships. (Take $\sqrt{3} \approx 1.732$).

Step-by-Step Solution:

1. **Draw the Diagram**:
2. Let AB be the lighthouse of height $h = 100\ \text{m}$.
3. Let C and D be the two ships on opposite sides on the sea level along a straight line.
4. The foot of the lighthouse is B.
5. Angle of elevation from ship C to top A is 45° (alternate interior angle to angle of depression).
6. Angle of elevation from ship D to top A is 60°.



1. **In right ΔABC**:

$$\tan 45^\circ = \frac{AB}{BC}$$

Since $\tan 45^\circ = 1$ and $AB = 100$:

$$1 = \frac{100}{BC} \implies BC = 100 \text{ m}$$

2. **In right ΔABD**:

$$\tan 60^\circ = \frac{AB}{BD}$$

Since $\tan 60^\circ = \sqrt{3}$ and $AB = 100$:

$$\sqrt{3} = \frac{100}{BD} \implies BD = \frac{100}{\sqrt{3}}$$

Rationalize the denominator:

$$BD = \frac{100\sqrt{3}}{3} = \frac{100 \times 1.732}{3} = \frac{173.2}{3} \approx 57.73 \text{ m}$$

3. **Total distance between the ships**:

$$\text{Distance } CD = BC + BD = 100 + 57.73 = \mathbf{157.73} \text{ m}$$

(Note: If ships were on the SAME side of the lighthouse, distance would be $BC - BD = 100 - 57.73 = 42.27$ m)

Template 2: Graphical Solution of Linear Equations (5 Marks — Q43)

Problem (Real Exam Question):

> Solve the following system of linear equations graphically:
>
> $$2x + y = 6$$
>
> $$2x - y = -2 \quad (\text{or } 2x - y + 2 = 0)$$

Step-by-Step Solution:

1. **Table of Values for Equation 1:** $2x + y = 6 \implies y = 6 - 2x$
2. If $x = 0 \implies y = 6 - 2(0) = 6 \implies (0, 6)$
3. If $x = 2 \implies y = 6 - 2(2) = 2 \implies (2, 2)$
4. If $x = 3 \implies y = 6 - 2(3) = 0 \implies (3, 0)$

x	0	2	3
y	6	2	0

1. **Table of Values for Equation 2:** $2x - y = -2 \implies y = 2x + 2$
2. If $x = 0 \implies y = 2(0) + 2 = 2 \implies (0, 2)$
3. If $x = 1 \implies y = 2(1) + 2 = 4 \implies (1, 4)$
4. If $x = -1 \implies y = 2(-1) + 2 = 0 \implies (-1, 0)$

x	0	1	-1
y	2	4	0

1. **Plotting on Graph Sheet**:
2. Plot points $(0, 6), (2, 2), (3, 0)$ and draw a straight line through them, labeling it $2x + y = 6$.
3. Plot points $(0, 2), (1, 4), (-1, 0)$ and draw a straight line through them, labeling it $2x - y = -2$.
4. **Point of Intersection**:

5. Both lines intersect at the point $P(1, 4)$.
6. **Conclusion**: The solution to the system is $\mathbf{x = 1,\ y = 4}$. *(Quick check:* $2(1) + 4 = 6$ ✓, $2(1) - 4 = -2$ ✓*)*

Template 3: Construction of Tangents to a Circle (3 to 5 Marks — Q39/Q43)

Problem:

> Draw a circle of radius $3\ \text{cm}$. From a point P at a distance of $7\ \text{cm}$ from the centre O, construct a pair of tangents to the circle. Measure the length of the tangents.

Step-by-Step Construction Steps (Write these exact steps in your answer sheet):

1. **Step 1**: Mark a point O as the centre and draw a circle of radius $3\ \text{cm}$ using a compass.
2. **Step 2**: Take a point P such that $OP = 7\ \text{cm}$. Join OP.
3. **Step 3**: Draw the perpendicular bisector of the line segment OP. Let M be the mid-point of OP ($OM = MP = 3.5\ \text{cm}$).
4. **Step 4**: With M as centre and radius MO (or MP), draw a circle that cuts the given circle at two points, say T_1 and T_2.
5. **Step 5**: Join PT_1 and PT_2.
6. PT_1 and PT_2 are the required pair of tangents from point P to the circle.

Mathematical Verification:

In right ΔOT_1P ($\angle OT_1P = 90^\circ$ because tangent $\perp$ radius):

$$PT_1 = \sqrt{OP^2 - OT_1^2} = \sqrt{7^2 - 3^2} = \sqrt{49 - 9} = \sqrt{40} \approx \mathbf{6.32}\ \text{cm}.$$

Template 4: Median of Grouped Frequency Distribution (3 Marks — Q42)

Problem (Real Exam Question):

> The distribution below gives the marks/weights of 30 students. Find the median:

Class Interval	$40-45$	$45-50$	$50-55$	$55-60$	$60-65$	$65-70$	$70-75$
Frequency (f)	2	3	8	6	6	3	2

Step-by-Step Solution:

1. **Prepare the Cumulative Frequency Table**:

Class Interval	Frequency (f)	Cumulative Frequency (cf)
$40-45$	2	2
$45-50$	3	$2+3=5$
$50-55$	8	$5+8=13$
$55-60$ **(Median Class)**	6 (f)	$13+6=19$
$60-65$	6	$19+6=25$
$65-70$	3	$25+3=28$
$70-75$	2	$28+2=30$
Total	$N=30$	

1. **Locate the Median Class**:

$$\frac{N}{2}=\frac{30}{2}=15$$

The cumulative frequency just greater than 15 is 19, which corresponds to the class interval $55 - 60$. Therefore, **Median Class** $= 55 - 60$.

2. **Identify the Formula Values**:

3. Lower limit of median class, $L = 55$
4. Class size, $h = 60 - 55 = 5$
5. Frequency of median class, $f = 6$

6. Cumulative frequency of preceding class, $cf = 13$

7. **Apply Median Formula**:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

$$\text{Median} = 55 + \left(\frac{15 - 13}{6} \right) \times 5 = 55 + \left(\frac{2}{6} \right) \times 5 = 55 + \frac{5}{3} = 55 + 1.67 = \mathbf{56.67}$$

Template 5: Coordinate Geometry Ratio & Section Formula (3 Marks — Q40)

Problem (Real Exam Question from 2025 Paper):

> Find the ratio in which the y-axis divides the line segment joining the points $A(4, -5)$ and $B(-1, 3)$. Also find the coordinates of the point of intersection.

Step-by-Step Solution:

1. **Set up the variables**:
2. Let the y-axis divide the segment AB in the ratio $k : 1$ at point P.

3. Any point on the y-axis has its x-coordinate equal to 0. So, let $P = (0, y)$.

4. **Apply Section Formula for the x-coordinate**:

$$x = \frac{k(x_2) + 1(x_1)}{k + 1}$$

Here $x_1 = 4,\ x_2 = -1,\ x = 0$:

$$0 = \frac{k(-1) + 1(4)}{k+1}$$

$$0 = -k + 4 \implies k = 4$$

Therefore, the required ratio is $k : 1 = \mathbf{4 : 1}$.

5. **Find the y-coordinate of P**:

$$y = \frac{k(y_2) + 1(y_1)}{k+1} = \frac{4(3) + 1(-5)}{4+1} = \frac{12-5}{5} = \frac{7}{5} = 1.4$$

6. **Conclusion**:

7. The ratio is $4 : 1$.
8. The coordinates of the point of intersection are $\left(\mathbf{0}, \frac{\mathbf{7}}{\mathbf{5}}\right)$.

Template 6: Circle Tangents Calculation (2 Marks — Q33)

Problem:

> From an external point P, two tangents PM and PN are drawn to a circle of radius $7\ \text{cm}$. If the distance of P from the centre O is $25\ \text{cm}$, find the length of each tangent.

Step-by-Step Solution:

1. Since the tangent to a circle is perpendicular to the radius through the point of contact, ΔOMP is a right-angled triangle with $\angle OMP = 90^\circ$.
2. By Pythagoras Theorem in right ΔOMP:

$$OP^2 = OM^2 + PM^2$$

$$25^2 = 7^2 + PM^2$$

$$625 = 49 + PM^2 \implies PM^2 = 625 - 49 = 576$$

$$PM = \sqrt{576} = \mathbf{24}\ \text{cm}$$

3. Since tangents from an external point are equal in length:

$$PN = PM = \mathbf{24}\ \text{cm}.$$